Characterising Stochastic Processes with Path Signatures

From One-Dimensional Moment Recovery to High-Dimensional Path Geometry and Advanced Financial Processes

PathLaw project

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Abstract

This report documents a staged study of how *truncated path signatures* characterise stochastic processes, together with the theory and implementation behind it. On the theory side we set out the stochastic differential equations (SDEs) and basic properties of the models we study — geometric Brownian motion, Heston stochastic volatility, the Merton and Bates jump-diffusions, fractional Brownian motion and rough Bergomi volatility — and give a self-contained account of path signatures, their truncation, and the precise sense in which the *expected signature* encodes the moments and law of a process. On the implementation side we detail how the two signature libraries (`pysiglib` and `roughpy`) were cross-validated, how the batched, memory-flat Monte-Carlo engine simulates each process, and how moments and signatures are estimated and compared. On the results side we confirm, on one-dimensional Gaussian and non-Gaussian processes, that each moment becomes visible at its own signature level (mean $\rightarrow$ 1, variance $\rightarrow$ 2, skewness $\rightarrow$ 3, kurtosis $\rightarrow$ 4); we recover cross-channel covariance geometry (level 2+) in high dimension and benchmark it against classical estimators; and we show that on advanced financial processes the signature separates stochastic volatility, jumps and, most distinctively, *path memory* (long memory and roughness) that a moment or covariance table cannot see. All signature summaries are streaming and verified flat at ≈ 0.46 GB for 100,000 paths.

Contents

1 Introduction

The guiding research question, posed by the project supervisor, is:

> *"We want to check the way signatures characterise processes by checking how they recover their moments. Note, you have to truncate the level of the signatures. Start with*

a Gaussian process (only two moments) and complexify the processes (higher moments) and test the truncation levels."

In plain terms: **can truncated path signatures characterise a stochastic process by recovering its moment information (mean, variance, ...), its geometry (covariance, co-movement) in higher dimensions, and its path memory once we leave the Gaussian/Markov world?**

Four departures from the Black–Scholes benchmark. The classical benchmark for an asset price is geometric Brownian motion (Black–Scholes): constant volatility and independent Gaussian log-returns. Real markets violate this in four qualitatively different ways, and these are the "cases" this report studies:

1. **Stochastic volatility.** Volatility is itself random and mean reverts, producing volatility clustering and (with leverage) skew. The canonical model is **Heston**.
2. **Jumps (and jump-diffusion).** Prices move discontinuously, creating skew and heavy tails on top of a diffusion. The canonical models are **Merton** (jumps on a constant-vol diffusion) and **Bates** (jumps on Heston stochastic vol — the combined jump-diffusion).
3. **Non-Markovian dynamics / long memory.** Increments carry autocorrelation, so the future depends on the whole past. The canonical driver is **fractional Brownian motion**.
4. **Rough volatility.** Volatility is not merely stochastic but *rough* — its trajectories are far less regular than Brownian motion and highly autocorrelated. The canonical model is **rough Bergomi**.

Part I (Theory) writes down the SDE and basic properties of each, then develops signatures and how they access moments. Part II (Implementation) details the library validation, the Monte-Carlo engine and the estimation/comparison machinery. Part III (Results) presents the experiments, staged from simple to advanced:

1. **Stage 1 (1D Gaussian).** Recover the mean and variance.
2. **Stage 2 (1D non-Gaussian).** Recover skewness and kurtosis with matched mean and variance.
3. **Stage 3 (high-dimensional Gaussian).** Recover cross-channel covariance geometry and benchmark against classical estimators.
4. **Stage 4 (advanced processes).** Separate the four departures above and locate each feature by signature level.

All signatures are computed with `pysiglib`; `roughpy` is used as an independent cross-check. Every experiment is synthetic, on data whose true moments and geometry are known, so the signatures can be checked against a known answer.

Part I
Theory

2 Stochastic models and their SDEs

We work throughout with the *log-price* $X_t = \log S_t$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$, with W, Z, W^S, W^v standard Brownian motions and N a Poisson process. All parameters are annualised and the horizon is T years.

2.1 The Black–Scholes benchmark (GBM)

Geometric Brownian motion is

$$\mathrm{d}S_t = \mu\, S_t\, \mathrm{d}t + \sigma\, S_t\, \mathrm{d}W_t, \qquad \Longleftrightarrow \qquad \mathrm{d}X_t = \Big(\mu - \tfrac{1}{2}\sigma^2\Big)\mathrm{d}t + \sigma\, \mathrm{d}W_t. \tag{1}$$

Properties. X has *independent, stationary, Gaussian* increments: $X_{t+\Delta} - X_t \sim \mathcal{N}\big((\mu - \tfrac{1}{2}\sigma^2)\Delta,\ \sigma^2\Delta\big)$. It is a Markov process, a (time-changed) Lévy process and, after discounting, a martingale. Its law is fixed by just two moments; skewness and excess kurtosis are zero. This is the null model against which every richer process is compared, and $\mathrm{Var}(X_T - X_0) = \sigma^2 T$ is the yardstick we match the other processes to.

2.2 Stochastic volatility: the Heston model

Heston [14] makes the variance v_t a mean-reverting square-root (Cox–Ingersoll–Ross [15]) process correlated with the price:

$$\mathrm{d}X_t = \Big(\mu - \tfrac{1}{2}v_t\Big)\mathrm{d}t + \sqrt{v_t}\, \mathrm{d}W_t^S, \tag{2}$$

$$\mathrm{d}v_t = \kappa(\theta - v_t)\, \mathrm{d}t + \xi\sqrt{v_t}\, \mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\, \mathrm{d}t. \tag{3}$$

Here κ is the mean-reversion speed, θ the long-run variance, ξ the vol-of-vol and ρ the leverage correlation. **Properties.** The pair (X, v) is jointly Markov, but X *alone* is not — it depends on the latent v. The variance stays positive when the *Feller condition* $2\kappa\theta \geq \xi^2$ holds (here $2 \cdot 2 \cdot 0.04 = 0.16 \geq 0.09 = 0.3^2$), and its stationary law is Gamma. Stochastic volatility produces *volatility clustering* (autocorrelated squared returns) and, through $\rho < 0$, negative skew and semi-heavy tails — excess kurtosis without the true jumps of the next model. Increments are *not* independent, so X is not a Lévy process.

2.3 Jumps: the Merton jump-diffusion

Merton [16] adds a compound-Poisson jump term to a constant-vol diffusion. With jump times of intensity λ and i.i.d. log jump sizes $J_i \sim \mathcal{N}(m, \delta^2)$,

$$X_t = X_0 + \Big(\mu - \tfrac{1}{2}\sigma^2 - \lambda\bar{\kappa}\Big)t + \sigma W_t + \sum_{i=1}^{N_t} J_i, \qquad \bar{\kappa} = \mathbb{E}\big[e^J - 1\big] = e^{m + \frac{1}{2}\delta^2} - 1, \tag{4}$$

where $-\lambda\bar{\kappa}$ is the martingale compensator that removes the drift the jumps would otherwise inject. **Properties.** X is a *Lévy process*: increments are independent and stationary, and it is Markov. Over a step Δ the increment has mean $(\mu - \frac{1}{2}\sigma^2 - \lambda\bar{\kappa})\Delta + \lambda m\Delta$, variance $\sigma^2\Delta + \lambda(m^2 + \delta^2)\Delta$, *negative skew* when $m < 0$, and *positive excess kurtosis* $\propto \lambda(3\delta^4 + 6m^2\delta^2 + m^4)$ — genuinely heavy (leptokurtic) tails absent from any diffusion.

2.4 Stochastic volatility with jumps: the Bates model

Bates [17] combines the two, superimposing Merton jumps on Heston stochastic volatility:

$$\mathrm{d}X_t = \Big(\mu - \tfrac{1}{2}v_t - \lambda\bar{\kappa}\Big)\mathrm{d}t + \sqrt{v_t}\,\mathrm{d}W_t^S + \mathrm{d}\Big(\textstyle\sum_{i=1}^{N_t} J_i\Big), \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \xi\sqrt{v_t}\,\mathrm{d}W_t^v. \tag{5}$$

Properties. It inherits volatility clustering and leverage from Heston and skew/heavy tails from Merton: the jumps dominate short-horizon tail behaviour while stochastic volatility shapes the term structure of higher moments. It is the richest of the diffusion-based models here and neither Markov in X alone nor Lévy.

2.5 Non-Markovian dynamics: fractional Brownian motion

Fractional Brownian motion (fBM) [20] B^H with Hurst index $H \in (0,1)$ is the centred Gaussian process with $B_0^H = 0$ and

$$\mathbb{E}\big[B_t^H B_s^H\big] = \tfrac{1}{2}\big(|t|^{2H} + |s|^{2H} - |t-s|^{2H}\big). \tag{6}$$

We take the log-price to be $X_t = c\,B_t^H$ (with c chosen so $\mathrm{Var}(X_T) = \sigma^2 T$, matching Black–Scholes). **Properties.** B^H is *self-similar*, $B_{at}^H \overset{d}{=} a^H B_t^H$, and has *stationary* increments. The increment process (fractional Gaussian noise, fGn) has autocovariance

$$\gamma(k) = \tfrac{1}{2}\big(|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H}\big), \tag{7}$$

so for $H > \frac{1}{2}$ increments are *positively correlated with long-range dependence* ($\sum_k \gamma(k) = \infty$, persistent/trending), for $H < \frac{1}{2}$ they are anti-persistent, and $H = \frac{1}{2}$ recovers Brownian motion. Crucially, for $H \neq \frac{1}{2}$ fBM is *neither Markov nor a semimartingale*: the future depends on the entire past. Its *marginal* law is still Gaussian, so moments alone cannot distinguish it from GBM once the variance is matched — only the path memory differs. We use $H = 0.70$.

2.6 Rough volatility: the rough Bergomi model

Empirically, log-volatility behaves like fBM with a *small* Hurst index $H \approx 0.1$ — "volatility is rough" [21]. The rough Bergomi model [22] drives the instantaneous variance by a rough Volterra (Riemann–Liouville) process $\widetilde{W}_t^H = \sqrt{2H}\int_0^t (t-s)^{H-\frac{1}{2}}\,\mathrm{d}Z_s$:

$$v_t = \xi_0\,\exp\Big(\eta\,\widetilde{W}_t^H - \tfrac{1}{2}\eta^2 t^{2H}\Big), \tag{8}$$

$$\mathrm{d}X_t = -\tfrac{1}{2}v_t\,\mathrm{d}t + \sqrt{v_t}\,\big(\rho\,\mathrm{d}Z_t + \sqrt{1-\rho^2}\,\mathrm{d}Z_t^{\perp}\big), \tag{9}$$

with forward variance ξ_0, vol-of-vol η, roughness $H \in (0, \frac{1}{2})$ and leverage ρ. The term $-\frac{1}{2}\eta^2 t^{2H}$ normalises $\mathbb{E}[v_t] = \xi_0$. **Properties.** Volatility paths have Hölder regularity $H - \varepsilon$ — far rougher than Brownian — and are strongly (anti-)autocorrelated at short lags. The variance is non-Markov and non-semimartingale; the model reproduces the steep short-maturity skew that classical stochastic-volatility models miss. We use $H = 0.10$, $\eta = 0.8$.

2.7 Summary of properties

Table 1: Qualitative properties of the six models. "Markov" refers to the observable log-price X alone; jump/rough/fractional models add structure that a two-moment description cannot capture.

Model	Markov in X	Increment law	Memory	Skew	Heavy tails
Black–Scholes	yes	Gaussian, i.i.d.	none	no	no
Heston	no (needs v)	non-Gaussian	clustering	yes (ρ)	mild
Merton	yes (Lévy)	Gaussian + jumps	none	yes ($m < 0$)	yes
Bates	no	jumps + SV	clustering	yes	yes
fBM (H=0.7)	no	Gaussian, correlated	long memory	no	no
rough Bergomi	no	non-Gaussian	rough/short memory	yes (ρ)	yes

3 Path signatures

A *path* is a continuous map $x : [0, T] \to \mathbb{R}^d$; discretely, a sequence of vectors. The *signature* [1, 2, 3] is the collection of all iterated integrals of the path. For a multi-index $(i_1, \dots, i_k) \in \{1, \dots, d\}^k$ the corresponding coordinate is

$$S(x)^{\,i_1\cdots i_k} = \int\cdots\int_{0<t_1<\cdots<t_k<T} \mathrm{d}x^{i_1}_{t_1}\,\mathrm{d}x^{i_2}_{t_2}\cdots\mathrm{d}x^{i_k}_{t_k} \;\in\; \mathbb{R}, \tag{10}$$

and the level-k tensor collects all d^k of them,

$$S(x)^{(k)} = \int\cdots\int_{0<t_1<\cdots<t_k<T} \mathrm{d}x_{t_1}\otimes\cdots\otimes\mathrm{d}x_{t_k} \;\in\; (\mathbb{R}^d)^{\otimes k}. \tag{11}$$

The full signature $S(x) = \big(1, S^{(1)}, S^{(2)}, \dots\big)$ lives in the tensor algebra $T((\mathbb{R}^d))$ and begins with the empty-word constant 1.

Two distinct axes. It is essential not to confuse two quantities:

Concept	Meaning	Range in this study
Path dimension d	number of channels / assets / factors	1 (or 2 with time), then 20/10
Signature level k	order of the iterated-integral feature	up to 3 or 5

At level k in dimension d there are exactly d^k coordinates, so the signature truncated at depth N has length $d + d^2 + \cdots + d^N$.

Algebraic structure. Two facts drive everything below. (i) *Chen's identity*: signatures are multiplicative under path concatenation, $S(x{*}y) = S(x)\otimes S(y)$, which is what makes them computable incrementally and streaming. (ii) The *shuffle identity*: the pointwise product of two signature coordinates is a linear combination (the shuffle) of higher signature coordinates. Consequently *every polynomial in the increments of the path is a linear functional of the signature.* This is the algebraic reason signatures give access to moments (Section 4).

3.1 Truncation and its implications

The signature is infinite, so in practice we **truncate** at a finite depth N, keeping levels $1, \dots, N$. Two properties make this benign and quantifiable:

- **Factorial decay.** For a path of length (1-variation) L, $\left\|S(x)^{(k)}\right\| \leq L^k/k!$ [3]. Higher levels shrink factorially, so a modest N already captures most of the analytic information — truncation error is controlled a priori.

- **Uniqueness.** The (full) signature determines a path up to tree-like equivalence and reparametrisation [5, 6]; adding a monotone *time channel* removes the reparametrisation ambiguity, so a time-augmented path is recovered from its signature. This is why we time-augment the one-dimensional processes in Stage 4.

The *implications* of truncation are the theme of this project: a depth-N signature can only resolve structure up to order N. A Gaussian needs $N = 2$; a skewed process needs $N \geq 3$; a heavy-tailed one needs $N \geq 4$. Truncating too shallow makes two processes that differ only in a higher moment look identical — exactly the phenomenon the experiments probe. (For dimensionality one may instead use the *log-signature*, a compressed basis of the free Lie algebra whose size is given by the Witt formula; we use the full signature throughout for simplicity, since our effective dimensions are small.)

3.2 The expected signature and characterisation of laws

For a *random* path X we form the **expected signature** $\Phi(X) = \mathbb{E}[S(X)]$, the average of the signature over the law of the path. Under moment-type conditions the expected signature *determines the law of the process* [7, 8, 10]: it is the path-space analogue of the classical moment problem, where the sequence of moments pins down a distribution. This is the precise sense in which "signatures characterise a process". Comparing two processes therefore reduces to comparing their expected signatures, and truncating at level N compares them up to N-th-order information — which is what we do, level by level, throughout the report. The signature method and these characterisation results are surveyed in [9]; efficient differentiable implementations and kernels are developed in [11, 12, 13].

4 Accessing moments with signatures

4.1 From signatures to moments (mathematically)

The one-dimensional case. For a scalar path X the level-k signature telescopes:

$$S(X)^{(k)} = \int \cdots \int_{0<t_1<\cdots<t_k<T} \mathrm{d}X_{t_1} \cdots \mathrm{d}X_{t_k} = \frac{\left(X_T - X_0\right)^k}{k!}. \tag{12}$$

Taking expectations, $\Phi(X)^{(k)} = \mathbb{E}\left[(X_T - X_0)^k\right]/k!$, so the level-$k$ coordinate of the expected signature is exactly the k-th *raw moment* of the total increment, divided by $k!$. Hence:

level 1 $\leftrightarrow$ mean, level 2 $\leftrightarrow$ 2nd moment, level 3 $\leftrightarrow$ 3rd moment (skew), level 4 $\leftrightarrow$ 4th moment (kurtosis).

This is why, in 1D, comparing expected signatures level by level recovers moment differences, and why higher 1D levels carry little *new* information (they are powers of a single number).

Time augmentation and path dependence. Augmenting to the two-dimensional path $t \mapsto (t, X_t)$ adds *cross* terms such as $\int_0^T t\,\mathrm{d}X_t$ and $\int_0^T X_t\,\mathrm{d}t$, which depend on the *order* in which the increments occur, not merely on the endpoint. These are what let the signature separate processes with identical endpoint moments but different memory — fractional and rough processes in Stage 4.

The multi-dimensional case. For $d > 1$ the level-2 term $\int \mathrm{d}X^i\,\mathrm{d}X^j$ splits into a symmetric part $\frac{1}{2}\,\Delta X^i \Delta X^j$ (whose expectation gives the *covariance*) and an antisymmetric part, the *Lévy area* $\frac{1}{2}(\int X^i \mathrm{d}X^j - \int X^j \mathrm{d}X^i)$, which encodes ordered co-movement a symmetric covariance matrix cannot. More generally, by the shuffle identity every mixed moment of the increments is a linear functional of the expected signature, so the truncated expected signature is a rich, ordered generalisation of the moment vector.

4.2 Monte-Carlo estimation of the process moments

Because the processes are simulated, their moments are estimated by Monte Carlo. From n i.i.d. paths we form the endpoint increments $\{\Delta X^{(p)}\}_{p=1}^n$ and estimate raw/central moments by sample averages: the sample mean $\hat{\mu} = \frac{1}{n}\sum_p \Delta X^{(p)}$, the sample variance, and standardised central moments for skewness and excess kurtosis. In the multivariate case we estimate the mean vector $\hat{\mu} = \frac{1}{n}\sum_p x^{(p)}$ and covariance $\widehat{\Sigma} = \frac{1}{n}\sum_p x^{(p)} x^{(p)\top} - \hat{\mu}\hat{\mu}^\top$. These are the "textbook" summaries a quant would reach for; their sampling error scales like $n^{-1/2}$, and *higher* moments are noisier, which is why Stage 2 uses more paths (a running sum of many increments is pushed toward Gaussianity by the central limit theorem, diluting the higher-moment signal).

4.3 Comparing moments and signatures

We compare processes under two lenses and report several normalisations.

- **Statistical distances.** L^2 distance between mean vectors and Frobenius distance between covariance matrices, $\|\hat{\mu}_A - \hat{\mu}_B\|_2$ and $\|\widehat{\Sigma}_A - \widehat{\Sigma}_B\|_F$.
- **Signature distances.** L^2 distance between expected signatures. Because a level-k block has d^k coordinates that are also numerically larger, a raw distance conflates "real difference" with "scale". We therefore report, per level: (1) the **raw** L^2 distance; (2) the **level-normalised** distance, dividing by $\sqrt{d^k}$ (a per-coordinate RMS that makes levels of different width comparable); and (3) the **standardised** distance, dividing each coordinate by the baseline process's signature-coordinate standard deviation before measuring, which removes the magnitude effect and exposes the true switch-on structure. The standardised view is the one we plot.

Part II
Implementation

5 Validation of the signature libraries

Signatures are easy to get subtly wrong: the truncation convention, the leading constant term, the channel ordering and the time parametrisation are all places a pipeline can silently disagree with the mathematics. We therefore cross-validate the two libraries before trusting any number.

Settings. Signatures are computed with `pysiglib` via `pysiglib.sig(paths, degree=`N`, time_aug, end_time, n_jobs=-1)`. The convention is `scalar_term=False`, i.e. the returned vector omits the leading 1 and is laid out as the concatenation $[\,\text{level }1\,|\,\text{level }2\,|\cdots|\,\text{level }N\,]$ with level k occupying d^k consecutive coordinates; `n_jobs` $= -1$ uses all cores. Arrays are passed as C-contiguous `float64` (else the library warns and copies). Time augmentation, when used, appends a single monotone channel on $[\,\text{end\_time}/T,\ \text{end\_time}\,]$.

Cross-check. We recompute the Stage-1 expected signatures at every level with both `pysiglib` and `roughpy`, matching all conventions (no scalar term, identical channel order, identical time parametrisation and depth). The two libraries agree to numerical precision: across all processes and levels the largest disagreement is $\approx 2 \times 10^{-13}$, i.e. floating-point rounding. This confirms the truncation layout and parametrisation before we scale up, and is the reason the level-indexing used by every distance table (Section 4.3) can be trusted.

6 The Monte-Carlo simulation engine

6.1 Streaming, batched design

No experiment ever stores all paths or all signatures at once. For each process we loop: (i) simulate a batch of paths (and their increments); (ii) update the classical estimators; (iii) compute the batch's signatures and fold them into fixed-length running sums; (iv) discard the batch. Memory is therefore independent of the number of paths — verified flat at ≈ 0.46 GB for a 100,000-path Stage-3 run. Reproducibility is guaranteed by seeding a dedicated `numpy.random.default_rng` per (configuration, process) with deterministic offsets (Python's built-in `hash` is randomised per process and is deliberately *not* used).

6.2 Discretisation schemes

Each model uses an explicit, documented scheme on a grid of `time_steps` points with step $\mathrm{d}t = T/$`time_steps`:

- **GBM / Merton diffusion:** exact Gaussian increments $(\mu - \frac{1}{2}\sigma^2)\mathrm{d}t + \sigma\sqrt{\mathrm{d}t}\,Z$.
- **Heston / Bates variance (CIR):** Euler *full-truncation* scheme [18, 19] $v_{t+1} = v_t + \kappa(\theta - v_t^+)\mathrm{d}t + \xi\sqrt{v_t^+}\sqrt{\mathrm{d}t}\,Z_v$ with the variance floored at 0 wherever it is used, and the correlated price innovation $Z_s = \rho Z_v + \sqrt{1-\rho^2}Z_\perp$.
- **Merton / Bates jumps:** per step draw $N_t \sim \text{Poisson}(\lambda \mathrm{d}t)$; the total log jump is $N_t\,m + \sqrt{N_t}\,\delta\,Z$ (a sum of N_t i.i.d. normals), less the compensator $\lambda\bar{\kappa}\,\mathrm{d}t$.
- **fBM:** exact fractional Gaussian noise via the Cholesky factor L of the Toeplitz covariance built from (7); a batch is $ZL^\top$. The factor depends only on (`time_steps`, H) and is cached across batches. The scale $\sigma\sqrt{T}/\texttt{time\_steps}^H$ matches the terminal variance to Black–Scholes.
- **Rough Bergomi:** the fractional process is built on the grid from the same Cholesky factor ($H = 0.1$); we form v_t by the martingale- normalised exponential and correlate the spot Brownian with the per-step vol innovation (leverage). Building the fractional process exactly in distribution while correlating the spot with the step innovation is a deliberate simplification of the full Volterra construction, adequate for this synthetic testbed.

The 1-D log-price is the cumulative sum of the increments; a monotone time channel is appended to the *path* (not the increments) when time augmentation is on.

6.3 Settings and parameters

Table 2: Global Monte-Carlo settings per stage (seed 42 throughout).

Stage	Paths	Steps	Horizon	Depth	Dimension
1 (1D Gaussian)	1,000	20	—	≤ 5	1
2 (1D moments)	4,000	20	—	≤ 5	1
3 (high-dim)	10,000	300	—	3 / 5	20 / 10
4 (advanced)	5,000	100	1 yr	5	1 (2 with time)

Table 3: Stage-4 model parameters (annualised; baseline volatility $\sigma = 0.2$, so $\sigma^2 = 0.04$). Batch size 500; time augmentation on.

Model	Parameters
Black–Scholes	$\sigma = 0.2,\ \mu = 0$
Heston	$v_0 = \theta = 0.04,\ \kappa = 2.0,\ \xi = 0.3,\ \rho = -0.7$
Merton	$\sigma = 0.2,\ \lambda = 1.0,\ m = -0.10,\ \delta = 0.15$
Bates	Heston + Merton parameters above
fBM	$H = 0.70$, scaled to $\mathrm{Var}(X_T) = \sigma^2 T$
rough Bergomi	$H = 0.10,\ \xi_0 = 0.04,\ \eta = 0.8,\ \rho = -0.7$

7 Computing and comparing moments with the libraries

7.1 Streaming expected signature

For each process a `RunningSignatureStats` accumulator holds three fixed-length quantities: the running sum of signature coordinates, the running sum of squares, and the path count. From these we recover, without ever materialising all signatures, the **expected signature** (sum / count) and the **per-coordinate standard deviation** (from $\mathrm{Var} = \mathbb{E}[x^2] - \mathbb{E}[x]^2$, clipped at 0), the latter being the baseline scale used for the standardised distances. At $d = 10$, depth 5 a single signature has 111,110 numbers (≈ 0.9 MB); storing all of them for 10,000 paths would need ≈ 8.9 GB, so this streaming is what makes the experiment feasible.

7.2 Streaming statistical estimators

The classical moments are estimated the same streaming way: running sums $\sum x$ and $\sum xx^\top$ (length d and $d \times d$) give the mean and covariance at the end via $\widehat{\Sigma} = \frac{1}{n}\sum xx^\top - \hat{\mu}\hat{\mu}^\top$. Terminal statistics use one sample per path (the endpoint); increment statistics use every $(\text{path}, \text{step})$ pair.

7.3 Distance measures in practice

Given the per-process expected signatures and statistical summaries, the runner builds three tables implementing Section 4.3: a statistical-distance table (L^2 on means, Frobenius on covariances), a whole-signature distance table (raw and level-normalised), and a level-wise table (raw, level-normalised and standardised by the baseline coordinate std). Because levels occupy disjoint coordinate blocks, the whole-signature distance truncated at depth k equals $\sqrt{\sum_{j \le k} (\text{raw level-}j\text{ distance})^2}$, so a single level table yields the distance-versus-depth curves without recomputation.

Part III
Results

8 Stage 1: one-dimensional Gaussian processes

We simulate three Gaussian random walks (1000 paths, 20 time steps, seed 42), changing one property at a time so each comparison isolates one moment.

Table 4: Stage 1 processes and the empirical moments of their increments. The simulator reproduces the intended mean and variance.

Process	True mean	True variance	Emp. mean	Emp. variance
A (baseline)	0.00	1.0	+0.005	1.009
B (mean shifted)	0.05	1.0	+0.045	1.007
C (volatility doubled)	0.00	4.0	+0.008	4.021

Comparisons: **A vs B** isolates the mean, **A vs C** isolates the variance, **B vs C** combines both.

Table 5: Stage 1 signature distances (L^2 between expected signatures) by truncation level. The mean difference is present from level 1; the variance difference switches on at level 2.

Comparison	L1	L2	L3	L4	L5
A_vs_B (mean)	0.808	0.808	7.29	9.16	34.7
A_vs_C (variance)	0.054	31.25	31.36	860.3	917.8
B_vs_C (combined)	0.862	31.25	32.79	854.8	901.5

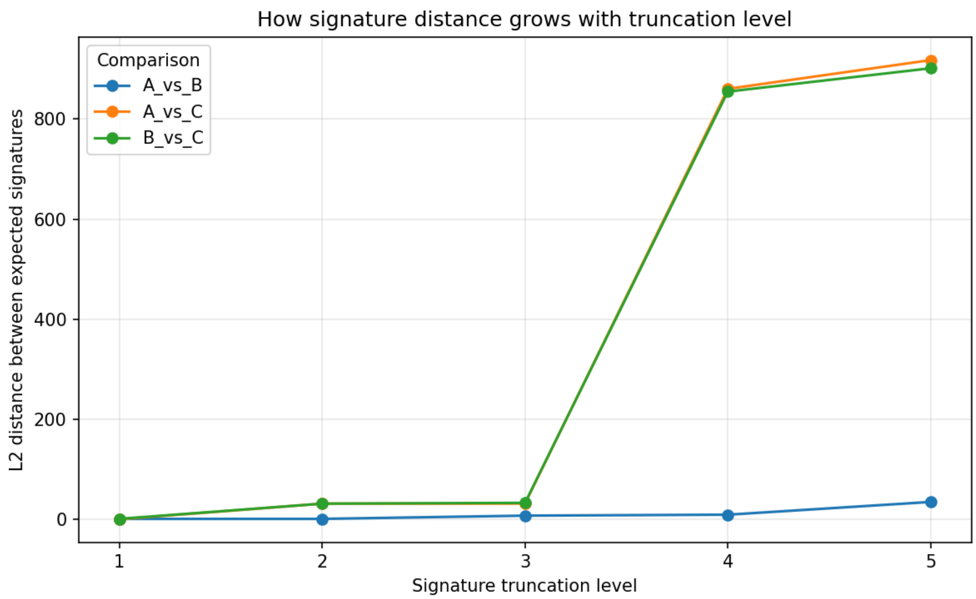

Figure 1: Stage 1: L^2 distance between expected signatures versus truncation level. The mean comparison (A_vs_B) is already non-zero at level 1; the variance comparison (A_vs_C) starts near zero and jumps at level 2.

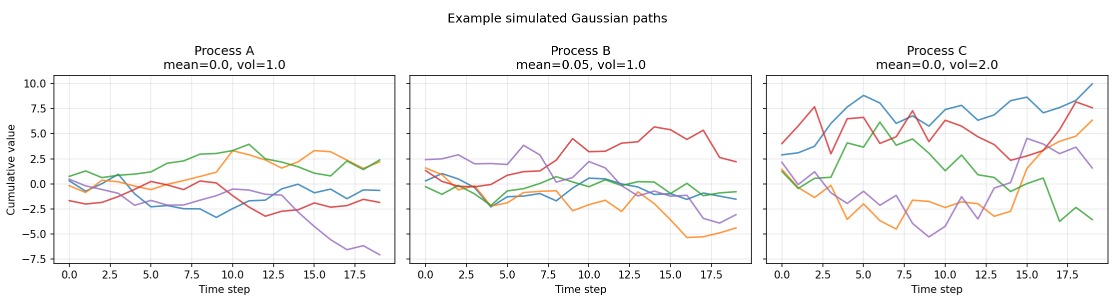

Figure 2: Example Stage 1 paths. Process C (double volatility) visibly wanders further; process B drifts gently upward.

Reading. The mean is a first-order (level-1) property, recovered immediately; the variance is a second-order (level-2) property, appearing only once level 2 is included. The continued growth of the raw distance at levels 4–5 is a *scale effect* (higher-level coordinates are numerically larger), not new distributional information — a Gaussian carries no independent information beyond its first two moments.

9 Stage 2: one-dimensional higher moments

Stage 2 matches the mean and variance across three processes so that levels 1–2 cannot separate them; any difference the signature detects must then come from a *higher* moment.

Table 6: Stage 2 processes (4000 paths). Mean and variance are matched; only skewness and kurtosis differ.

Process	Mean	Variance	Skewness	Excess kurtosis
G (Gaussian)	-0.00	1.01	$+0.01$	$+0.03$
S (skew-normal, $\alpha = 8$)	-0.00	0.99	$+$**0.92**	$+0.75$
T (Student-t, df $= 5$)	-0.00	1.00	$+0.05$	$+$**4.44**

Table 7: Stage 2 signature distances by level. Skewness (G_vs_S) switches on at level 3; kurtosis (G_vs_T) switches on at level 4.

Comparison	L1	L2	L3	L4	L5
G_vs_S (skewness)	0.011	0.119	**1.781**	6.089	21.83
G_vs_T (kurtosis)	0.033	0.033	0.749	**6.006**	9.914
S_vs_T (combined)	0.022	0.127	1.036	1.045	13.12

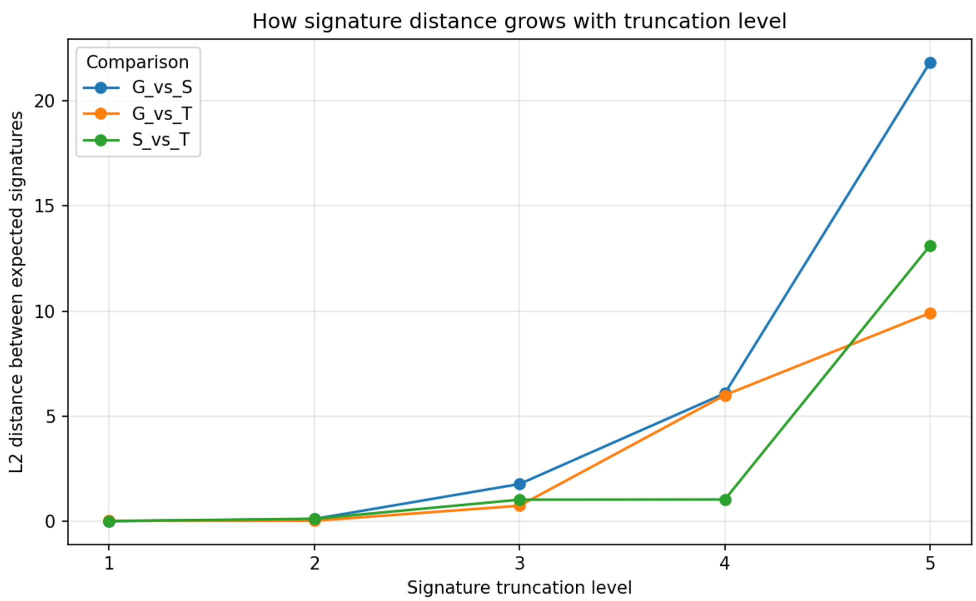

Figure 3: Stage 2: distance versus level. The skew line (G_vs_S) switches on at level 3; the heavy-tail line (G_vs_T) stays small through level 3 (the Student-t is symmetric) and switches on at level 4.

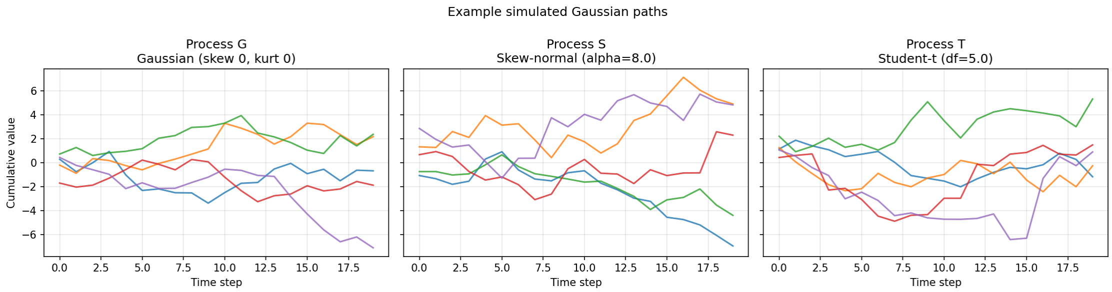

Figure 4: Example Stage 2 paths for G, S and T.

Reading. Each moment becomes visible at its own signature level: mean → 1, variance → 2, skewness → 3, kurtosis → 4. To "see" the k-th moment one must truncate the signature at level $\geq k$ — a direct demonstration of the supervisor's point that the truncation level matters and that richer processes need deeper truncation.

10 Stage 3: high-dimensional Gaussian benchmark

10.1 Motivation: from moments to geometry

A one-dimensional path has no geometry: the signature essentially sees only the endpoint, and higher levels behave like powers of a single number (Eq. (12)). Once a path has $d > 1$ channels, level 2 onward contains genuine cross-channel terms $S^{i,j}$ (and ordered higher-order interactions $S^{i,j,k}, \dots$), which encode **covariance, correlation blocks and co-movement**. Stage 3 tests this and compares signatures against classical statistical estimators.

Path meaning. Each path is a multivariate return stream $X_t = (r_t^1, \dots, r_t^d)$, where r_t^i is the increment of *synthetic asset* i. Thus $d = 20$ means 20 synthetic assets evolving together, and $d = 10$ means 10.

10.2 Configurations and processes

Because the feature count grows as d^k, we use two complementary settings.

Table 8: The two configurations and their signature feature dimensions ($d + d^2 + \cdots + d^{\text{depth}}$, no constant term).

Config	Purpose	d	depth	Feature dim	Batch size
`d20_depth3`	wide, shallow: cross-channel geometry	20	3	8,420	250
`d10_depth5`	narrow, deep: higher-order interactions	10	5	111,110	100

All four processes are multivariate Gaussian random walks with increments $\mathrm{d}X \sim \mathcal{N}(\mu\,\mathrm{d}t,\ \Sigma\,\mathrm{d}t)$, $\mathrm{d}t = 1$.

Table 9: The four high-dimensional processes.

Process	Mean structure	Covariance structure	Interpretation
A baseline	0	identity	d independent assets, no drift, unit vol
B mean-shift	first 25% of channels = 0.05	identity	a subset of assets has a directional trend
C corr.-block	0	identity + correlated blocks	sector / block co-movement
D common-shock	0	market factor + idiosyncratic	a market-wide factor drives co-movement

The correlation blocks in C set pairwise correlation $\rho = 0.5$ on channels 0–4 and $\rho = 0.3$ on channels 5–9 (remaining channels independent). Because only one property changes per process, each comparison isolates one effect: A vs B (mean), A vs C and A vs D (covariance geometry), B vs C (mixed).

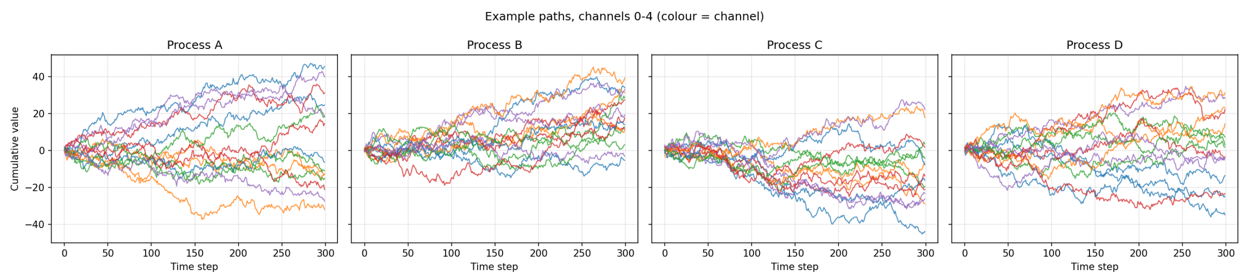

Figure 5: Example high-dimensional paths, channels 0–4 (colour = channel), one panel per process. Process C's low channels visibly co-move (correlation blocks); Process D shows market-wide co-movement.

10.3 Results: classical statistical distances

Table 10: Statistical distances between processes (10,000 paths, 300 steps). The mean-shift (B) is seen only by the mean estimator; the correlation block (C) and common shock (D) are seen only by the (increment) covariance estimator. The increment covariance is the clean detector: the terminal covariance scales with the number of steps and so has a large sampling-noise floor.

Config	Comparison	Mean-vec L^2	Incr. cov. Frob.	Term. cov. Frob.
`d20_depth3`	A_vs_B (mean)	**33.35**	0.017	82.5
`d20_depth3`	A_vs_C (corr)	1.00	**2.610**	791.6
`d20_depth3`	A_vs_D (shock)	1.09	**1.798**	532.7
`d10_depth5`	A_vs_B (mean)	**21.33**	0.009	42.9
`d10_depth5`	A_vs_C (corr)	1.02	**2.612**	773.8
`d10_depth5`	A_vs_D (shock)	0.88	**0.901**	280.5

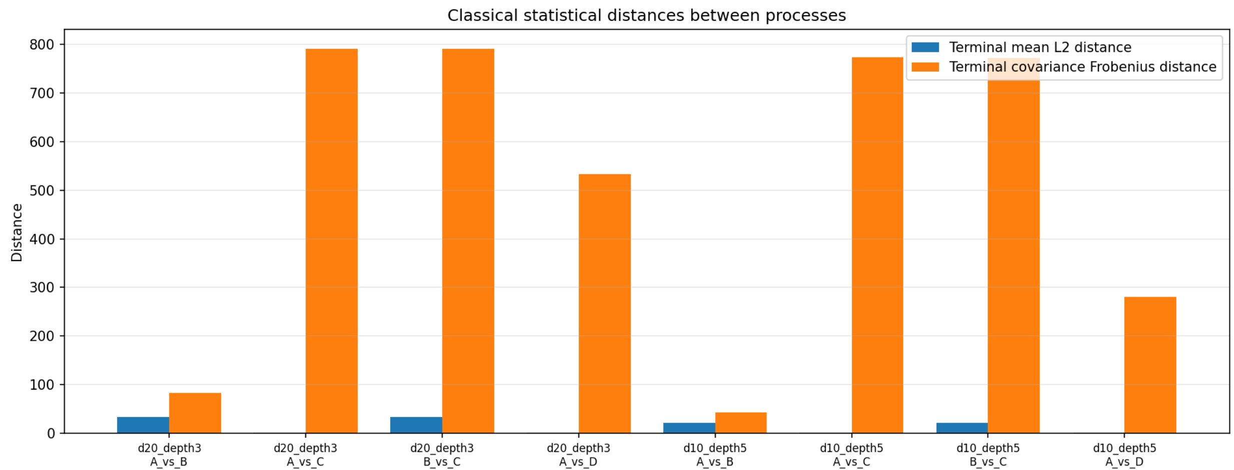

Figure 6: Classical statistical distances. The mean-shift comparisons have small covariance distance; the correlation/shock comparisons have large covariance distance and near-zero mean distance.

10.4 Results: signature distances

Table 11: Standardised level-wise signature distances (each coordinate in units of Process A's variability). The mean shift (A_vs_B) peaks at level 1; the covariance-geometry comparisons (A_vs_C, A_vs_D) are near-zero at level 1 and peak at level 2.

Config	Comparison	L1	L2	L3	L4	L5
`d20_depth3`	A_vs_B	**0.433**	0.135	0.084	—	—
`d20_depth3`	A_vs_C	0.013	**0.094**	0.014	—	—
`d20_depth3`	A_vs_D	0.014	**0.064**	0.015	—	—
`d10_depth5`	A_vs_B	**0.388**	0.109	0.098	0.033	0.028
`d10_depth5`	A_vs_C	0.018	**0.183**	0.015	0.074	0.016
`d10_depth5`	A_vs_D	0.016	**0.068**	0.015	0.028	0.016

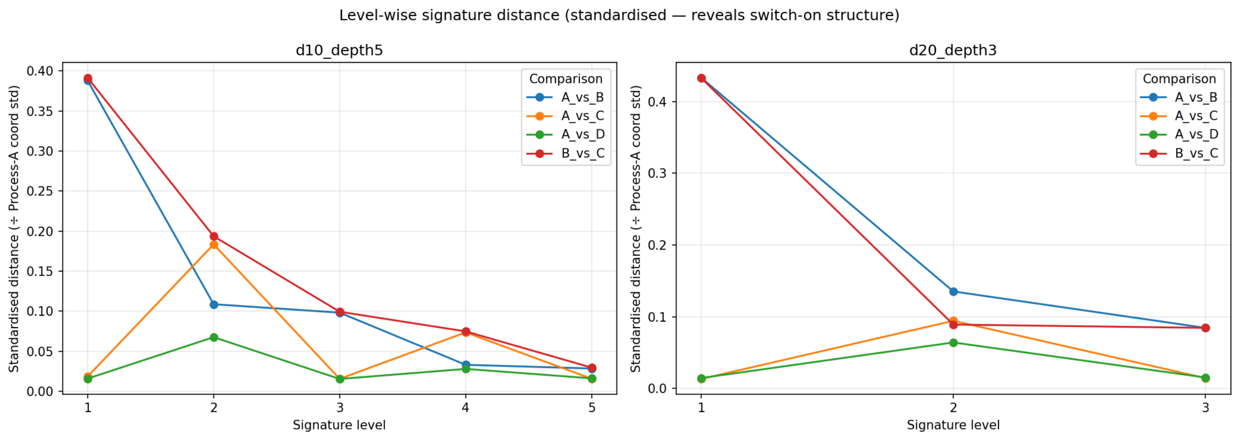

Figure 7: The key figure: standardised level-wise signature distance. A_vs_B (mean shift) peaks at level 1; A_vs_C and A_vs_D (covariance geometry) are flat at level 1 and peak at level 2. The mixed comparison B_vs_C is elevated at both.

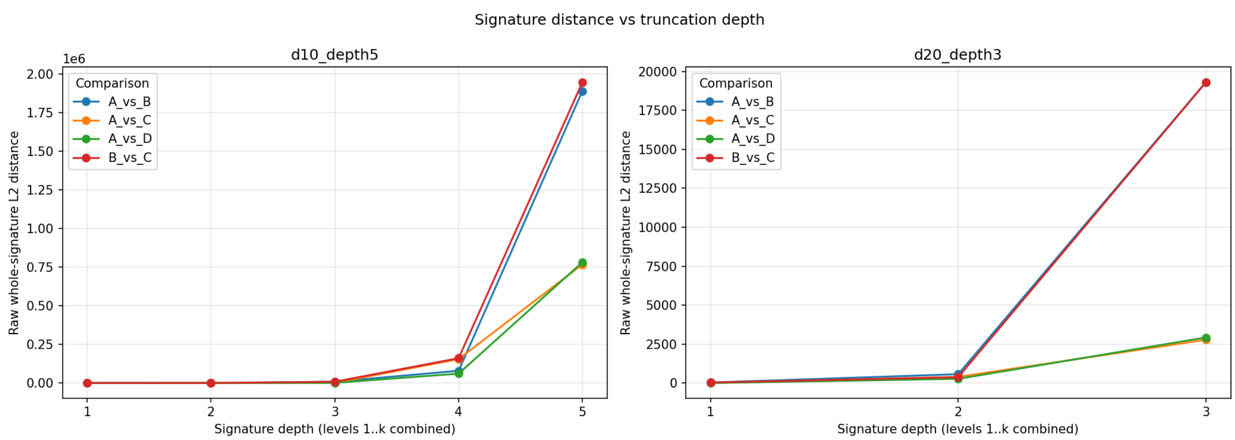

Figure 8: Raw whole-signature distance as depth grows (levels 1..k combined). The raw distance is dominated by the magnitude of the highest included level — a scale effect — which is exactly why the standardised view above is needed to read the true structure.

The scale caveat, made concrete. The raw and level-normalised distances grow monotonically with level for *every* comparison. For instance, the raw A_vs_C level-3 distance in `d20_depth3` is ≈ 2748, dwarfing the level-2 value of ≈ 396 — purely a magnitude artifact. Only after standardising by Process A's coordinate standard deviation does the genuine "level-2 covariance" signal emerge. Normalised or standardised distances must therefore always be reported alongside the raw ones.

10.5 Scaling

A full run of both configurations at 100,000 paths peaked at ≈ 0.46 GB resident memory — flat regardless of path count, as the streaming design intends — and took ≈ 2 min for `d20_depth3` and ≈ 26 min for `d10_depth5`. The extra paths only sharpen the results: the A_vs_C level-1 noise floor falls from 0.013 at 10,000 paths to 0.004 at 100,000, making the level-2 covariance signal stand out even more clearly.

11 Stage 4: advanced financial processes

Stage 4 takes the six models of Section 2 — Black–Scholes plus the four departures — as 1-D log-price paths, all calibrated to the same baseline volatility so that levels 1–2 are comparable and the interesting differences live in the higher moments and the path memory. Time augmentation is on (effective dimension 2), so the signature is sensitive to path ordering; at depth 5 the feature vector has only $2+4+8+16+32 = 62$ coordinates.

Table 12: Stage-4 processes and empirical statistics (5,000 paths, 100 steps, horizon 1 year, seed 42). All share the baseline volatility; the differences are in the higher moments and the path memory.

Code	Process (feature)	Term. var.	Incr. var.	Term. mean $\|\cdot\|$
`BS`	Black–Scholes (baseline)	0.0401	0.000401	0.021
`HEST`	Heston (stochastic vol)	0.0421	0.000401	0.021
`MERT`	Merton (jumps)	0.0718	0.000727	0.029
`BATES`	Bates (jump-diffusion)	0.0736	0.000727	0.034
`FBM`	fractional BM (H=0.70, memory)	0.0387	**0.000063**	0.002
`RVOL`	rough Bergomi (H=0.10, rough vol)	0.0644	0.000427	**0.297**

`BS` attains its designed variance $\sigma^2 T = 0.04$. Jumps roughly double the variance and shift the mean (negative-skew jumps). `FBM` has a much *smaller* per-step increment variance yet the *same* terminal variance: its positively correlated increments accumulate to the same endpoint spread — the defining fingerprint of long memory. `RVOL`'s large negative mean (the $-\frac{1}{2}v\,\mathrm{d}t$ Itô term under a heavy-tailed variance) reflects rough, spiky volatility.

Table 13: Standardised level-wise signature distances from the Black–Scholes baseline (each coordinate in units of `BS`'s variability). Bold marks where each feature switches on: stochastic vol grows through the higher levels; jumps switch on at level 2; Bates is the strongest of the diffusion family; long memory (`FBM`) is modest but non-zero despite a matched terminal variance; rough volatility dominates at every level.

Comparison	L1	L2	L3	L4	L5
`BS_vs_HEST` (stoch. vol)	0.001	0.020	0.092	0.134	**0.191**
`BS_vs_MERT` (jumps)	0.040	**0.276**	0.373	0.449	0.545
`BS_vs_BATES` (jump-diff.)	0.065	**0.295**	0.417	0.519	0.646
`BS_vs_FBM` (long memory)	**0.114**	0.073	0.135	0.171	0.188
`MERT_vs_BATES` (added vol)	0.025	0.026	0.090	0.119	0.168
`BS_vs_RVOL` (rough vol)	1.367	1.266	1.587	1.991	**2.608**
`FBM_vs_RVOL` (memory kinds)	1.481	1.327	1.650	2.043	**2.650**

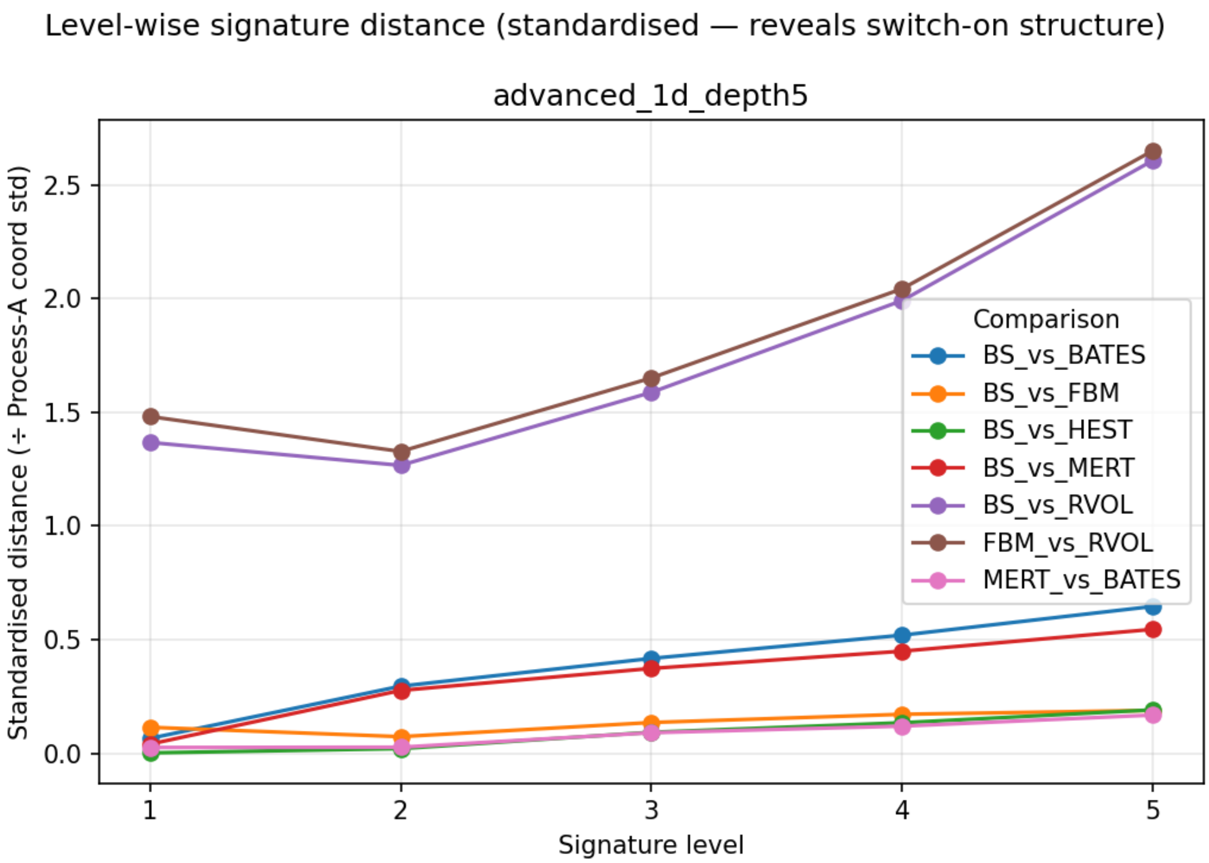

Figure 9: Stage 4: standardised level-wise signature distance from Black–Scholes. Rough volatility (`RVOL`) is the most extreme departure at every level; below it, jumps (`MERT`, `BATES`) switch on at level 2 while stochastic vol (`HEST`) and long memory (`FBM`) contribute smaller, higher-order signals. (The axis label "÷ Process-A coord std" refers to the `BS` baseline, reusing the Stage-3 plotting code.)

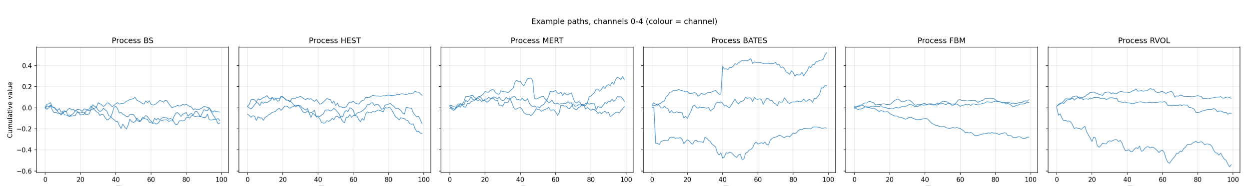

Figure 10: Example Stage-4 log-price paths, one panel per process. Jump processes show discrete drops; the rough-vol path is visibly the most erratic.

The blind spot of moments, made concrete. The classical increment-covariance distance between `BS` and `FBM` is only $\approx 3.4 \times 10^{-4}$ — essentially the same as any other pair — because fractional Gaussian noise still has a well-defined marginal variance. The moments are therefore nearly *blind* to `FBM`'s long memory. Yet the signature still separates `FBM` from `BS` (raw distance 0.029, larger than the stochastic-vol and jump pairs). This is the clearest single illustration in the whole study of what signatures add over a moment/covariance table: they encode *ordered,* *path-dependent* structure that summary statistics cannot see.

12 Discussion

Across all four stages the signature behaves as the theory predicts, and in the high-dimensional and advanced settings it agrees with the classical estimators while adding structure a moment table alone cannot order:

- **Mean shift** (A vs B) is a level-1 effect, recovered by both the mean estimator and signature level 1.
- **Covariance geometry** (A vs C correlation blocks; A vs D common shock) is a level-2+ effect carried by the cross terms $S^{i,j}$, recovered by both the covariance estimator and signature level 2.
- **Higher levels amplify scale**, so only normalised / standardised distances reveal the true switch-on structure.
- The signature additionally encodes *ordered* interactions ($S^{i,j} \neq S^{j,i}$), a form of path geometry that symmetric covariance matrices cannot express.
- **Advanced processes** (Stage 4) confirm the pattern beyond the Gaussian world: stochastic volatility and jumps appear as higher-moment effects (jumps switching on at level 2), while *path memory* — long memory in `FBM` and roughness in `RVOL` — is separated by the signature even when the marginal moments are matched and the classical increment-covariance estimator is blind to it.

13 Conclusion and next steps

Truncated path signatures characterise these processes by recovering their moment and geometric information at interpretable levels: mean $\to$ 1, variance $\to$ 2, skewness $\to$ 3, kurtosis $\to$ 4, and multi-channel covariance / co-movement $\to$ 2+. Beyond the Gaussian world, they further separate stochastic volatility, jumps and jump-diffusions, and — most distinctively — the *path memory* of fractional Brownian motion and rough volatility, which the marginal moments cannot see. All experiments are fully streaming and scale to 100,000 paths at constant memory.

Deliberately out of scope for now (the natural next stages) are: applying the pipeline to the real `SPY/QQQ/TLT` rolling windows; and signature kernels and signature-based anomaly / regime detection.

Reproducibility. All results use seed 42. The experiments are run from the project root with:

```
python "Gaussian testing/run_gaussian_signature_test.py"
python "Gaussian testing/run_higher_moment_test.py"
python "HighDim testing/run_highdim_gaussian_benchmark.py"
python "Advanced processes testing/run_advanced_processes_test.py"
```

Tables and figures are written under each folder's `results/` directory.

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